

Internal memorialization note

Selector sufficiency and outer-frame descent

in Type-1, $N = 3$ Landau-Zener

Proof record from the present conversation

This note memorializes two proofs developed in the conversation:

1. the *generic simple local selector sufficiency theorem*;
2. the *outer-frame descent theorem after resolved-sheet lift and half-form regularization*.

It is written to be detailed enough that a technically prepared reader can recreate the arguments without having to reconstruct the conversational steps. It also records the numerical assays that were useful in shaping and stress-testing the theorem statements.

Important scope statement. The descent theorem proved here is the intrinsic one, i.e. after passing to the resolved-sheet coordinate $u = u_* + t^2$ and packaging amplitudes as half-forms. In that formulation the earlier unresolved gauge bridge collapses. This is stronger than the older numerically supported picture, but it is also more precise about the normalization being used.

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1 Executive summary

At the April 18 baseline, the local Type-1 program already had the common cover, the commuting triple (H_x, Y, R) , canonical gluing on the regular set, the candidate selector sieve, and the exact two-sheet torus theorem, but still listed selector sufficiency and descent back to the original outer matcher as unresolved items; the same supplement treated outer descent through two narrow numerically supported assumptions and explicitly left selector sufficiency open [8, 7, 4]. The conversation then supplied two advances:

1. a proof that the older bridge ambiguity disappears once the local problem is written intrinsically on the resolved-sheet t -disk with half-form packaging;
2. a proof that every simple phase-active transport ramification point is an actual local selector point in the canonical two-sheet model.

What this note proves.

Outer-frame descent (intrinsic form). Let v be a simple transport ramification point. After resolved-sheet lift and half-form regularization, the quotient-ring/adiabatic exact local frame and the canonical resolved-sheet exact frame differ by a single constant geometry-only matrix G_v on the full t -disk; with matched normalization one has $G_v = I$.

Selector sufficiency (generic simple local form). Let v be a simple transport ramification point and fix a Borel direction θ . If the local cubic phase coefficient satisfies $e^{-i\theta}\sigma_v(H) \in \mathbb{R}^\times$, then the corresponding local wall is a genuine Airy Stokes wall, and crossing it inserts the exact local selector block

$$J_{v,\theta}^{\text{loc}}(t; H) = I + s_v^{\text{BS}}(t; H)E_{21}$$

or its opposite triangular companion, with nonzero scalar multiplier

$$s_v^{\text{BS}}(t; H) = -i e^{2\Theta_v(t; H)} \neq 0.$$

Hence every simple phase-active transport ramification point is an actual local selector point in the canonical two-sheet normalization.

2 Background and previously established inputs

2.1 Standing Type-1 notation

We work in the Type-1, $N = 3$ setting with the standard rational parametrization

$$u(\lambda) = \frac{n(\lambda)}{p(\lambda)}, \quad E(\lambda) = \frac{m(\lambda)}{p(\lambda)}, \quad q_u(\lambda) = u p(\lambda) - n(\lambda), \quad (2.1)$$

and quotient-ring fiber

$$\mathcal{E}_u = \mathbb{C}[\lambda]/(q_u(\lambda)). \quad (2.2)$$

The gauged adiabatic evolution is

$$f'_i(u) = - \sum_{j \neq i} S_{ij}(u) f_j(u) + i(E_i(u) - E_x(u)) f_i(u), \quad (2.3)$$

while the quotient-ring reduction is

$$\partial_u[R] = i[(m - E_x p) s_u R] = i H_x[R], \quad p s_u \equiv 1 \pmod{q_u}. \quad (2.4)$$

These objects, together with the commuting triple (H_x, Y, R) and the common/spinor cover picture, are formalized in the primer and strategy diagram [3, 4, 1].

2.2 Simple transport ramification points

The transport quartic is

$$W_4(\lambda) = n(\lambda)p'(\lambda) - n'(\lambda)p(\lambda) = p(\lambda)^2 \Sigma^{(2)}(\lambda). \quad (2.5)$$

A *simple transport ramification point* is a point $v \in \mathbb{P}_\lambda^1$ such that

$$W_4(v) = 0, \quad W'_4(v) \neq 0, \quad u_* = u(v). \quad (2.6)$$

Then the base map $u = n/p$ has simple critical behavior at v , and on the local double cover we may write

$$u = u_* + t^2. \quad (2.7)$$

The two active local inverse branches are denoted $\lambda_\pm(t)$, and the spectator sheet is omitted from the active 2×2 local analysis.

2.3 Exact two-sheet torus theorem

At a simple ramification point, the reduced sheet carriers satisfy exact scalar equations

$$\partial_u R_\pm = i(E_\pm - E_x) R_\pm. \quad (2.8)$$

On the t -cover this becomes the exact diagonal system

$$\partial_t \begin{pmatrix} R_+ \\ R_- \end{pmatrix} = (A_v(t; H)I + B_v(t; H)\sigma_3) \begin{pmatrix} R_+ \\ R_- \end{pmatrix}. \quad (2.9)$$

Reconstructing the half-form amplitudes gives

$$\Psi_{\pm}(t; H) = \sqrt{2t} \Gamma_{\pm}(t) R_{\pm}(t; H) \sqrt{dt}. \quad (2.10)$$

The April 18 theorem states that after removing the common scalar phase, the exact local matching gauge in the canonical resolved-sheet / WKB normalization is exactly torus-valued:

$$\hat{G}_v(t; H) = \exp(\Theta_v(t; H) \sigma_3). \quad (2.11)$$

Equivalently, no torus-breaking off-diagonal term survives in that canonical two-sheet normalization [7, 8].

2.4 Local Airy normalization and the candidate selector condition

The local phase mismatch between the two colliding sheets is

$$S_{+-}(u; H) = \sigma_v(H)(u - u_*)^{3/2} + O((u - u_*)^{5/2}), \quad (2.12)$$

where $\sigma_v(H) \neq 0$ in the generic simple case. The local selector notes computed the linear Langer/Airy coordinate

$$\zeta_v(H; u) = \left(\frac{3i}{4} \sigma_v(H) \right)^{2/3} (u - u_*), \quad (2.13)$$

which rectifies the leading cubic Airy exponent, and exhibited an explicit wall/on-wall member together with an off-wall control [5]. The normalized Airy jump for an $S_1 \rightarrow S_0$ crossing is

$$J_{1 \rightarrow 0}^{\text{norm}} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} = I - E_{21}. \quad (2.14)$$

In the canonical WKB normalization, the exact local selector multiplier is scalar:

$$s_v^{\text{BS}}(t; H) = -i \exp(2\Theta_v(t; H)), \quad (2.15)$$

so the exact local block is a scalar renormalization of the universal Airy jump [7, 8, 6].

2.5 What was still open before the conversation proofs

The older local matching note identified one unresolved bridge matrix,

$$Y_{v,s}^{qa}(u) = Y_v^{rs}(u) G_{v,s}(H), \quad (2.16)$$

connecting the resolved-sheet basis to the original quotient-ring/adiabatic frame, and therefore left the exact local jump in the original frame in the form

$$J_{v;s \rightarrow s'}^{qa}(H) = G_{v,s'}(H)^{-1} J_{s \rightarrow s'}^{\text{norm}} G_{v,s}(H). \quad (2.17)$$

At the same April 18 baseline, selector sufficiency and the exact global insertion rule were still listed among the remaining symbolic tasks [5, 8, 7].

3 Item 3: outer-frame descent after resolved-sheet lift and half-form regularization

3.1 Statement

The theorem we prove is the intrinsic bridge theorem. It is deliberately formulated after resolved-sheet lift and half-form regularization, because in that formulation the local bridge becomes exact and sector-independent.

Theorem 3.1 (Intrinsic outer-frame descent / bridge lemma). *Let v be a simple transport ramification point and write $u = u_* + t^2$. Let $L_+ \oplus L_-$ be the ordered active resolved-sheet splitting on a sufficiently small t -disk D_t . Let $Y_v^{rs}(t; H)$ denote the canonical exact resolved-sheet frame in that ordered splitting, after removing the common scalar phase. Let $Y_v^{qa}(t; H)$ denote the exact frame obtained by taking the quotient-ring carriers, reconstructing the adiabatic amplitudes, and packaging them as half-forms on the same t -disk.*

Then there exists a constant matrix $G_v \in GL_2(\mathbb{C})$ such that

$$Y_v^{qa}(t; H) = Y_v^{rs}(t; H) G_v \quad \text{for all } t \in D_t. \quad (3.1)$$

Moreover:

1. G_v is sector-independent, because it is defined on the whole connected t -disk;
2. G_v is geometry-only, i.e. it does not depend on the chosen commuting-family member H ;
3. if the two frames are matched by the same half-form normalization, then $G_v = I$ identically.

Consequently the old sector-labeled bridge $G_{v,s}(H)$ disappears in the intrinsic half-form formulation, and the exact jump in the quotient-ring/adiabatic half-form frame is just one fixed conjugate of the canonical local Airy jump.

3.2 Proof

The proof has three ingredients: the exact diagonal active carrier system, the regularity of the half-form package on the t -disk, and the identification of the quotient-ring/adiabatic half-form frame with the canonical resolved-sheet exact frame.

Lemma 3.2 (Regular half-form factors). *At a simple transport ramification point, the functions*

$$T_{\pm}(t) := \sqrt{2t} \Gamma_{\pm}(t) \quad (3.2)$$

are holomorphic and nonzero at $t = 0$.

Proof. Expand $\Sigma^{(2)}(\lambda)$ along the two local branches $\lambda_{\pm}(t)$. Because $W_4(v) = 0$ and $W'_4(v) \neq 0$, one has

$$\Sigma_{\pm}(t) := \Sigma^{(2)}(\lambda_{\pm}(t)) = \pm A_1 t + A_2 t^2 \pm A_3 t^3 + A_4 t^4 + O(t^5) \quad (3.3)$$

with $A_1 \neq 0$ [6]. Since $\Gamma_{\pm} = \delta_{\pm} \Sigma_{\pm}^{-1/2}$, we obtain

$$T_{\pm}(t) = \sqrt{2} \delta_{\pm} t^{1/2} \Sigma_{\pm}(t)^{-1/2}.$$

The factor $\Sigma_{\pm}(t)$ has a simple zero at $t = 0$, so $t^{1/2} \Sigma_{\pm}(t)^{-1/2}$ is holomorphic and nonzero there. Hence $T_{\pm}$ is holomorphic and nonzero at $t = 0$. $\square$

Lemma 3.3 (Active half-form frame is holomorphic on the full t -disk). *After half-form packaging, the active exact local system has no residual sectorial singularity at $t = 0$.*

Proof. The exact active carrier system (2.9) is diagonal with analytic coefficients on the t -disk. Therefore the carrier fundamental matrix

$$F_v(t; H) := \text{diag}(R_+(t; H), R_-(t; H)) \quad (3.4)$$

is holomorphic on D_t . Multiplying by the holomorphic nonvanishing factors $T_{\pm}(t)$ from Lemma 3.2 gives the half-form packaged exact frame

$$Y_v^{hf}(t; H) := \text{diag}(T_+(t), T_-(t)) F_v(t; H). \quad (3.5)$$

Thus the active exact system is represented by a holomorphic fundamental matrix on the entire disk. The apparent square-root singularity has been removed by the resolved-sheet lift and half-form packaging. $\square$

Proof of Theorem 3.1. Start from the exact diagonal carrier system (2.9). The quotient-ring/adiabatic

reconstruction formula is sheetwise and diagonal:

$$f_{\pm}(t; H) = \Gamma_{\pm}(t) R_{\pm}(t; H). \quad (3.6)$$

After packaging as half-forms using $\sqrt{du} = \sqrt{2t} \sqrt{dt}$, the resulting local frame is exactly

$$Y_v^{qa}(t; H) = \text{diag}(\sqrt{2t} \Gamma_+(t), \sqrt{2t} \Gamma_-(t)) F_v(t; H), \quad (3.7)$$

up to the common scalar phase that is removed in determinant-one normalization.

Now compare with the canonical resolved-sheet exact frame. In the April 18 two-sheet theorem the exact resolved-sheet half-form amplitudes are written as

$$Y_v^{\text{exact}}(t; H) = \text{diag}(T_+(t) e^{\Phi_{\text{rel}}(t; H)}, T_-(t) e^{-\Phi_{\text{rel}}(t; H)}), \quad (3.8)$$

again after removing the common scalar phase [7, 8]. But this is precisely the same construction as (3.7): the carrier matrix F_v is diagonal, its determinant-one part is $\text{diag}(e^{\Phi_{\text{rel}}}, e^{-\Phi_{\text{rel}}})$, and the half-form prefactors are $T_{\pm} = \sqrt{2t} \Gamma_{\pm}$. Hence, with matched normalization,

$$Y_v^{qa}(t; H) = Y_v^{rs}(t; H) \quad \text{for all } t \in D_t. \quad (3.9)$$

This already gives $G_v = I$.

If one uses two different but still exact normalizations of the same local horizontal frame, let the resulting frames be denoted Y_v^{qa} and Y_v^{rs} . Both are holomorphic fundamental matrices of the same active exact connection on the simply connected disk D_t . Therefore

$$\partial_t((Y_v^{rs})^{-1} Y_v^{qa}) = 0, \quad (3.10)$$

so their ratio is constant:

$$(Y_v^{rs})^{-1} Y_v^{qa} = G_v.$$

Because the two frames preserve the ordered splitting $L_+ \oplus L_-$, this constant is actually diagonal; determinant-one normalization places it in the torus. In any case it is sector-independent because it is defined on the entire connected disk.

Finally, G_v is geometry-only. In matched normalization $G_v = I$, so this is immediate. In a different normalization, the only possible difference comes from constant rescalings attached to the ordered lines $L_{\pm}$ and the choice of half-form convention; those are fixed by the local resolved-sheet geometry and do not depend on the commuting-family member. The H -dependence lives in the carrier phases and therefore cancels in the ratio.

This proves (3.1) and all three claims. $\square$

Corollary 3.4 (Intrinsic replacement for the older descent assumptions). *In the intrinsic resolved-*

sheet / half-form formulation, the older raw outer-matcher assumptions

1. no nondecaying torus-breaking off-diagonal survives after spectator decoupling and curvature subtraction;
2. the residual constant non-torus part is sector-independent and geometry-only,

are no longer independent hypotheses. They are absorbed by the exact identity (3.9) (or, in a different exact normalization, by the constant geometry-only bridge (3.1)).

Remark 3.5 (Scope of the descent theorem). The theorem is exact in the intrinsic half-form formulation. It does *not* claim that an arbitrary pre-regularized “raw outer matcher” is automatically identical without first being expressed in the same resolved-sheet / half-form gauge. The conversation proof deliberately moved to that intrinsic normalization because that is where the bridge becomes exact rather than only numerically supported.

4 Item 1: selector sufficiency in the generic simple local case

4.1 Statement

Theorem 4.1 (Generic simple local selector sufficiency). *Let v be a simple transport ramification point and fix a Borel direction θ . Assume:*

1. $W_4(v) = 0$ and $W'_4(v) \neq 0$;
2. the local phase mismatch is nondegenerate,

$$S_{+-}(u; H) = \sigma_v(H)(u - u_*)^{3/2} + O((u - u_*)^{5/2}), \quad \sigma_v(H) \neq 0; \quad (4.1)$$

3. the local disk contains no other transport ramification point and no ordinary turning point.

Then, in the intrinsic resolved-sheet / half-form formulation:

1. if $e^{-i\theta}\sigma_v(H) \in \mathbb{R}^\times$, the corresponding local wall through u_* is an actual selector boundary;
2. crossing that wall once produces the exact local jump

$$J_{v,\theta}^{\text{loc}}(t; H) = I + s_v^{\text{BS}}(t; H)E_{21} \quad (4.2)$$

or the opposite triangular companion, depending on orientation, with nonzero scalar multiplier

$$s_v^{\text{BS}}(t; H) = -i e^{2\Theta_v(t; H)} \neq 0; \quad (4.3)$$

3. therefore every simple phase-active transport ramification point is an actual local selector point

in the canonical two-sheet model.

4.2 Proof

The proof is local and has four steps: exact analytic rectification of the singulant, identification of the phase-active wall with a true Airy Stokes wall, nontriviality of the universal Airy jump, and invariance of nontriviality under diagonal torus conjugation.

Lemma 4.2 (Odd analyticity of the local singulant on the t -cover). *On the resolved-sheet cover $u = u_* + t^2$, the exact local phase mismatch $S_{+-}(t; H)$ is an odd analytic function of t .*

Proof. The two active local branches satisfy $\lambda_-(t) = \lambda_+(-t)$, because they are exchanged by the deck involution of the simple transport ramification. The phase differential on the common/spinor cover is $\omega = \sqrt{y} du$ [3, 4]. Exchanging the two colliding sheets reverses the sign of the sheetwise phase difference, so the local action mismatch changes sign under $t \mapsto -t$:

$$S_{+-}(-t; H) = -S_{+-}(t; H).$$

Since the branch-resolved data are analytic on the t -disk, $S_{+-}(t; H)$ is analytic there. Hence it is an odd analytic function of t . $\square$

Lemma 4.3 (Exact nonlinear Langer coordinate). *Assume $\sigma_v(H) \neq 0$. Define*

$$\Xi_{\theta,v}(t; H) := \frac{3}{4} e^{-i\theta} i S_{+-}(t; H). \quad (4.4)$$

Then there exists a holomorphic function $\chi_{\theta,v}(u; H)$ near u_ such that*

$$e^{-i\theta} i S_{+-}(u; H) = \frac{4}{3} \chi_{\theta,v}(u; H)^{3/2} \quad (4.5)$$

exactly, and

$$\chi_{\theta,v}(u; H) = \kappa_{\theta,v}(H)(u - u_*) + O((u - u_*)^2), \quad \kappa_{\theta,v}(H) = \left(\frac{3}{4} e^{-i\theta} i \sigma_v(H) \right)^{2/3}. \quad (4.6)$$

Proof. By Lemma 4.2 and the nondegenerate expansion (4.1), one has

$$\Xi_{\theta,v}(t; H) = c_{\theta,v}(H) t^3 (1 + t^2 a(t; H)), \quad c_{\theta,v}(H) \neq 0, \quad (4.7)$$

with $a(t; H)$ analytic. Choose a branch of $c_{\theta,v}(H)^{2/3}$ and define

$$\chi_{\theta,v}(t; H) := \Xi_{\theta,v}(t; H)^{2/3} = c_{\theta,v}(H)^{2/3} t^2 (1 + t^2 a(t; H))^{2/3}. \quad (4.8)$$

The factor $(1 + t^2 a)^{2/3}$ has a holomorphic branch near $t = 0$, so $\chi_{\theta,v}(t; H)$ is holomorphic. Because only even powers of t occur, it descends to a holomorphic function of $u - u_* = t^2$; we denote the

descended function by the same symbol $\chi_{\theta,v}(u; H)$. By construction,

$$\Xi_{\theta,v}(t; H) = \chi_{\theta,v}(u; H)^{3/2},$$

which is exactly (4.5). The linear term (4.6) follows from the leading cubic coefficient. $\square$

Proof of Theorem 4.1. Step 1: *The exact local system is already diagonal and torus-valued.* By the exact two-sheet theorem (2.11), the canonical exact local gauge is

$$\widehat{G}_v(t; H) = e^{\Theta_v(t; H)\sigma_3}.$$

Therefore the only possible local jump is a diagonal conjugate of the universal Airy jump. In particular, diagonal conjugation can rescale a scalar multiplier, but it cannot create or destroy the presence of a nontrivial triangular jump.

Step 2: *Phase-activity identifies a genuine Airy Stokes wall.* By Lemma 4.3, the exact singulant is the pullback of the Airy cubic exponent:

$$e^{-i\theta} i S_{+-}(u; H) = \frac{4}{3} \chi_{\theta,v}(u; H)^{3/2}.$$

Thus the local Stokes geometry is exactly the pullback of the standard Airy Stokes geometry under the holomorphic map $\chi_{\theta,v}$. The phase-active condition

$$e^{-i\theta} \sigma_v(H) \in \mathbb{R}^\times \tag{4.9}$$

means precisely that the linear part $\kappa_{\theta,v}(H)$ in (4.6) aligns the candidate wall through u_* with one of the canonical Airy Stokes rays. Therefore the local wall singled out by transport ramification plus phase resonance is not merely formal: it is an actual Stokes wall of the exact rectified Airy model.

Step 3: *Across that wall, the universal Airy jump is nontrivial.* In the normalized Airy model, for an $S_1 \rightarrow S_0$ crossing one has the universal jump (2.14). This is nontrivial. The opposite orientation gives the opposite triangular companion; in either case the model jump is rank-one triangular and nonzero [5].

Step 4: *Diagonal torus conjugation preserves nontriviality and yields the exact scalar block.* Let

$$D_v(t; H) := e^{\Theta_v(t; H)\sigma_3} = \text{diag}(d_+(t; H), d_-(t; H)).$$

Then

$$J_{1 \rightarrow 0}^{\text{loc}}(t; H) = D_v(t; H)^{-1} J_{1 \rightarrow 0}^{\text{norm}} D_v(t; H) = I - \frac{d_+(t; H)}{d_-(t; H)} E_{21}. \tag{4.10}$$

In the WKB-normalized Airy convention of the April 18 notes, the scalar coefficient is written as

$$s_v^{\text{BS}}(t; H) = -i e^{2\Theta_v(t; H)},$$

so the exact local block has the form (4.2). Since the exponential never vanishes, $s_v^{\text{BS}}(t; H) \neq 0$. Thus the exact local jump is genuinely nontrivial.

This proves that every simple phase-active transport ramification point inserts a nonzero local selector block in the canonical two-sheet model. The result is local, so it is independent of any global continuation word. Hence the candidate selector point is an actual local selector point. $\square$

Corollary 4.4 (Off-wall control on the same geometric ray). *Let H_{off} be a nearby commuting-family member for which $e^{-i\theta}\sigma_v(H_{\text{off}}) \notin \mathbb{R}$. Then the same geometric wall is not mapped by $\chi_{\theta,v}(\cdot; H_{\text{off}})$ to a canonical Airy Stokes ray. Consequently the normalized Airy basis continues analytically across that same wall for H_{off} , and the exact diagonal gauge cannot create a jump there.*

Proof. If $e^{-i\theta}\sigma_v(H_{\text{off}}) \notin \mathbb{R}$, the linear coefficient $\kappa_{\theta,v}(H_{\text{off}})$ in (4.6) does not send the chosen wall to one of the fixed Airy Stokes rays. Since the local exact geometry is the pullback of the Airy geometry under $\chi_{\theta,v}$, no Airy Stokes crossing occurs on that wall. The exact local gauge is diagonal, so there is no independent mechanism left to create a jump. $\square$

Remark 4.5 (Transfer to the quotient-ring/adiabatic half-form frame). By Theorem 3.1, the same local statement holds in the quotient-ring/adiabatic half-form frame: the local block is transferred by the constant geometry-only conjugacy G_v , and with matched normalization $G_v = I$.

5 Reconstruction blueprint: how to recreate the proofs

This section gives an explicit recipe for reconstructing the two proofs from the underlying formalism.

5.1 Blueprint for the intrinsic outer-frame descent theorem

To recreate Theorem 3.1, proceed in the following order.

1. Start from the quotient-ring reduction (2.4) and restrict to a simple transport ramification point v .
2. Lift to the resolved-sheet coordinate $u = u_* + t^2$ so that the two active branches are analytic and the carrier system takes the exact diagonal form (2.9).
3. Expand $\Sigma^{(2)}(\lambda_{\pm}(t))$ to obtain (3.3). This shows that $T_{\pm} = \sqrt{2t}\Gamma_{\pm}$ is regular and nonzero at $t = 0$.
4. Form the half-form packaged quotient-ring/adiabatic frame (3.7).
5. Compare it with the canonical exact resolved-sheet frame (3.8). In matched normalization they are identical. In any other exact normalization, differentiate the ratio to obtain (3.10) and conclude constancy.

6. Observe that any residual constant comes only from linewise normalization conventions and is therefore geometry-only.

5.2 Blueprint for the generic simple local sufficiency theorem

To recreate Theorem 4.1, proceed as follows.

1. Start from the local singulant expansion (4.1) at a simple transport ramification point.
2. Pass to the resolved-sheet coordinate t and use the deck involution to show that $S_{+-}(t; H)$ is odd analytic.
3. Build the exact nonlinear Langer coordinate by defining $\Xi_{\theta,v} = \frac{3}{4}e^{-i\theta}iS_{+-}$ and then $\chi_{\theta,v} = \Xi_{\theta,v}^{2/3}$. The odd-to-even structure ensures that χ descends holomorphically to the u -plane.
4. Check the phase-active condition $e^{-i\theta}\sigma_v(H) \in \mathbb{R}^\times$. This guarantees that the chosen local wall is the pullback of a canonical Airy Stokes wall.
5. Insert the universal Airy jump (2.14) and conjugate it by the exact diagonal torus gauge (2.11). This yields the exact local selector block (4.2) with nonzero scalar multiplier (4.3).
6. For an off-wall control, repeat the same calculation but note that the chosen wall no longer maps to an Airy Stokes wall. Therefore no jump is inserted there.

6 Numerical assays that were useful in validating these theorems

The proofs above are symbolic and do not require numerical input. The numerics were nevertheless important for three reasons: (i) they motivated the older outer-matcher conjectures, (ii) they stress-tested the selector sufficiency premise in the extreme-curvature regime, and (iii) they provided a reproducible audit trail that the local statements survived large random samples. Table 1 records the assays most relevant to the two theorems proved in this note.

Table 1: Numerical assays useful in shaping the theorem statements.

Assay	Precision	Accepted samples	Headline result	Primary relevance
Resolved-sheet off-diagonal probe	80 digits	1200	No off-diagonal counterexample found: 1200/1200 “no”	Supports exact two-sheet torus structure

Assay	Precision	Accepted samples	Headline result	Primary relevance
Spectator-decoupled torus survival	80 digits	1001 (including 301 targeted high-curvature)	1001/1001 pass; all 9/9 legacy high-curvature failures pass on retest	Older outer-descent picture
Sector-independence of residual constant part	80 digits base, 100 digits refined retest	1001 (including 301 targeted high-curvature)	999/1001 base passes; 2 borderline misses both pass refined 100-digit retest	Older outer-descent picture
Lagrange-inversion phase-recurrence validation	80 digits	1200	1200/1200 pass at 10^{-2} and 1198/1200 at 10^{-3} for (D_1, D_3, D_5) simultaneously	Validates scalar local exponent inputs
Selector-sufficiency random assay	60 digits	1302 (1001 generic + 301 targeted high-curvature)	1302/1302 pass; no accepted counterexamples; wall residual max 1.30×10^{-8} ; off-wall same-ray residual min 1.262×10^{-3}	Stress-tests local sufficiency premise
Selector-sufficiency replay / audit	70 digits	1302 replayed	1302/1302 recomputed passes; no replay failures; only tiny numerical differences from original log	Reproducibility audit

6.1 Assays inherited from the April 18 supplement

The April 18 supplement and main text recorded the numerical route that originally led to the older outer-matcher canonical form: 1200/1200 resolved-sheet off-diagonal “no” results, 1001/1001 torus-survival passes including 301 targeted high-curvature cases, 999/1001 base sector-independence passes with both misses clearing 100-digit retests, and 1200/1200 or 1198/1200 recurrence-validation passes depending on threshold [8, 7]. Those data were decisive in establishing that the descent question was likely narrow rather than structural.

6.2 Selector sufficiency assay developed in the conversation

The conversation added a dedicated random-sample assay of the local sufficiency premise. The operational statement tested was:

For a simple transport ramification point v , if a commuting-family member is tuned to the wall $\text{Im } \sigma_v(H) = 0$ at $\theta = 0$, then the exact local action mismatch S_{+-} places a genuine Stokes boundary at the Airy-predicted selector ray; a matched off-wall control does not.

The logged 60-digit run accepted 1302/1302 samples and found no accepted counterexamples. The replay script then recomputed the same diagnostics on the exact same 1302 logged samples at 70 digits and again found 1302/1302 passes, with only tiny replay-vs-log deviations in the derived diagnostics.

The most useful headline numbers were:

$$\begin{aligned} \text{wall residual threshold} &= 10^{-6}, & \text{off-wall same-ray threshold} &= 10^{-3}, \\ \max(\text{wall_relres}) &= 1.299372954971326 \times 10^{-8}, \\ \min(\text{off_rel_same_ray}) &= 1.262239463557905 \times 10^{-3}, \\ \min(\text{off_shift_rad}) &= 8.414931995034863 \times 10^{-4}, \\ \max(\text{curvature proxy}) &\approx 9.6118170775 \times 10^5. \end{aligned}$$

The replay reported

$$\begin{aligned} \max(\text{wall_relres}) &= 1.3139151918755005 \times 10^{-8}, \\ \min(\text{off_rel_same_ray}) &= 1.262239463625 \times 10^{-3}, \\ \max |\Delta \sigma_{\text{wall}}| &= 3.0762841493005714 \times 10^{-8}, \\ \max |\Delta \sigma_{\text{off}}| &= 1.214867814500203 \times 10^{-7}, \end{aligned}$$

with no replay failures. These numbers came directly from the machine-readable summaries and replay report generated in the conversation.

6.3 Companion assay artifacts generated in the conversation

The following files were created in the conversation and are the direct machine-readable support for the selector sufficiency assay and its replay.

- `selector_sufficiency_assay_log.xlsx`
- `selector_sufficiency_assay_accepted.csv`
- `selector_sufficiency_assay_borderline.csv`
- `selector_sufficiency_assay_counterexamples.csv`
- `selector_sufficiency_assay_summary.json`
- `selector_sufficiency_assay_report.txt`

- `selector_sufficiency_assay_replay.py`
- `selector_sufficiency_assay_replay_driver.sh`
- `selector_sufficiency_assay_replay_results.csv`
- `selector_sufficiency_assay_replay_diffs.csv`
- `selector_sufficiency_assay_replay_summary.json`
- `selector_sufficiency_assay_replay_report.txt`
- `selector_sufficiency_assay_replay_combined.log`

The replay artifacts are especially useful because they provide an explicit audit trail from code to recomputed results.

7 Status after these two proofs

Excluding nongeneric collision scenarios, these two proofs remove the two principal generic local uncertainties that remained at the April 18 baseline: the local bridge to the quotient-ring/adiabatic half-form frame and the local selector sufficiency statement. What remains is the exact *global* insertion rule: proving that crossing an active selector point inserts the scalar local block into the continuation functor in the correct active $2 \oplus 1$ embedding, with the spectator line untouched and with no radius/overlap ambiguity. That was already the single broken morphism $j_{H,\theta}$ in the formal strategy diagram, and it remains the next independent symbolic task [4].

8 Conclusion

The conversation proofs can be summarized in one sentence:

after resolved-sheet lift and half-form regularization, the local bridge is exact and constant, and once the exact local singulant is rectified to Airy form, simple phase-active transport ramification points necessarily carry nonzero scalar selector blocks.

This leaves the global insertion problem – not the local bridge, not the local Airy reduction, and not the local activation criterion in the generic simple case – as the next theorem-level frontier.

References

- [1] *Multistate Landau-Zener Solver: Augmented ODE with Gauged Dynamics* (internal summary).

- [2] *A note on the sextic gap polynomial for $N = 3$ Type-1 matrices* (internal note).
- [3] *Type-1, $N = 3$ Landau-Zener primer: double cover, reduced ODE, gluing theorem, and selector* (internal primer).
- [4] *Formalized strategy diagram for Type-1, $N = 3$ Landau-Zener* (internal note).
- [5] *Local selector matching at a transport ramification point in Type-1, $N = 3$* (internal note, April 16, 2026).
- [6] *Transport versus phase in the local gauge expansion at a simple transport ramification point* (internal note, April 17, 2026).
- [7] *All-orders torus structure and affine-linear local phase recurrences in Type-1, $N = 3$ Landau-Zener* (main text draft, April 18, 2026).
- [8] *Technical Supplement: all-orders torus structure, phase recurrence, and numerical support* (supplement draft, April 18, 2026).
- [9] *Single-kernel transport progress* (conversation memorandum and companion derivation, April 19, 2026).
- [10] `selector_sufficiency_assay_report.txt`, `selector_sufficiency_assay_summary.json`, `selector_sufficiency_assay_log.xlsx`, and companion CSV artifacts generated in the conversation.
- [11] `selector_sufficiency_assay_replay.py`, `selector_sufficiency_assay_replay_report.txt`, `selector_sufficiency_assay_replay_summary.json`, `selector_sufficiency_assay_replay_combined.csv`, and companion replay CSV artifacts generated in the conversation.